Python Exception Handling (Theory + Code)

1. Introduction

Exception handling allows programs to handle errors gracefully without crashing.

2. Basic try-except

```
try:
    print(10 / 0)
except ZeroDivisionError:
    print("Cannot divide by zero")
```

3. Multiple Exceptions

```
try:
    num = int(input())
    print(10 / num)
except ZeroDivisionError:
    print("Divide by zero")
except ValueError:
    print("Invalid input")
```

4. Else Block

```
try:
    num = int(input())
    result = 10 / num
except ZeroDivisionError:
    print("Error")
else:
    print("Result:", result)
```

5. Finally Block

```
try:
    f = open("file.txt", "r")
except FileNotFoundError:
    print("File not found")
finally:
    print("Execution finished")
```

6. Raising Exceptions

```
age = int(input())
if age < 18:
    raise Exception("Not eligible")
```

7. Custom Exception

```
class MyError(Exception):
    pass

raise MyError("Custom error")
```

8. File Handling Example

```
try:
    with open("data.txt", "r") as file:
        print(file.read())
except FileNotFoundError:
    print("File missing")
```

9. Backend Example

```
def login(user):
    if user != "admin":
        raise Exception("User not found")
    return "Success"

try:
    print(login("admin"))
except Exception as e:
    print(e)
```

10. Common Exceptions

ZeroDivisionError, ValueError, TypeError, FileNotFoundError, IndexError, KeyError

11. Practice Programs

```
# safe division
try:
    a = int(input())
    b = int(input())
    print(a / b)
except:
    print("Error")
```